EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

1. Mean of the first n terms of the A.P. $a, (a+d), (a+2d)$, ... is :

(A) $a+\frac{nd}{2}$ (B) $a+\frac{(n-1)d}{2}$ (C) $a+(n-1)d$ (D) $a+nd$

MSI066

2. The mean of 9 terms is 15. If one new term is added and mean become 16, then the value of new term is:

(A) 23 (B) 25 (C) 27 (D) 30

MSI067

3. If the mean of first n natural numbers is equal to $\frac{n+7}{3}$, then n is equal to:

(A) 10 (B) 11 (C) 12 (D) none of these

MSI068

4. The mean of first three terms is 14 and mean of next two terms is 18. The mean of all the five terms is :

(A) 15.5 (B) 15.0 (C) 15.2 (D) 15.6

MSI069

5. The mean of a set of numbers is $\bar{x}$. If each number is decreased by λ, then the mean of the new set is :

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(A) $\bar{x}$ (B) $\bar{x}+\lambda$ (C) $\lambda-\bar{x}$ (D) $\bar{x}-\lambda$

MSI070

6. The mean of 50 observations is 36. If its two observations 30 and 42 are deleted, then the mean of the remaining observations is :

(A) 48 (B) 36 (C) 38 (D) none of these

MSI071

7. The A.M. of n observations is $\bar{x}$. If the sum of $n-4$ observations is K, then the mean of remaining observations is :

(A) $\frac{\bar{x}-K}{4}$ (B) $\frac{n\bar{x}-K}{n-4}$ (C) $\frac{n\bar{x}-K}{4}$ (D) $\frac{n\bar{x}-(n-4)K}{4}$

MSI072

8. The mean of two samples of sizes 20 and 10 were found to be 11, 8 respectively. Their variance were 4 and 34 respectively. Find the variance of combined sample of size 30.

(A) 19 (B) 19.5 (C) 18.5 (D) 16

MSI014

9. The weighted mean of first n natural numbers when their weights are equal to corresponding natural number is:

(A) $\frac{n+1}{2}$ (B) $\frac{2n+1}{3}$ (C) $\frac{(n+1)(2n+1)}{6}$ (D) None of these

MSI073

For More Join: @IITJEE_Advanced

10. In a group of students, the mean weight of boys is 65 kg and the mean weight of girls is 55 kg. If the mean weight of all the students of group is 61 kg, then the ratio of the number of boys and girls in the group is :

(A) 2 : 3 (B) 3 : 1 (C) 3 : 2 (D) 4 : 3

MSI074

11. The median of the numbers 6, 14, 12, 8, 10, 9, 11 is :

(A) 8 (B) 10 (C) 10.5 (D) 11

MSI075

12. Median of the following frequency distribution

x_i	3	6	10	12	7	15
f_i	3	4	2	8	13	10

is :

(A) 7 (B) 10 (C) 8.5 (D) None of these

MSI076

13. Median is independent of change of :

(A) Only origin (B) Only scale

(C) Origin and scale both (D) Neither origin nor scale

MSI077

14. A series which have numbers three 4's, four 5's, five 6's, eight 7's, seven 8's and six 9's, then the mode of numbers is :

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(A) 9 (B) 8 (C) 7 (D) 6

MSI078

15. Mode of the following frequency distribution

x_i	4	5	6	7	8
f_i	6	7	10	8	3

is :

(A) 5 (B) 6 (C) 8 (D) 10

MSI079

16. The mode of the following frequency distribution is :

Class	1-10	11-20	21-30	31-40	41-50
f_i	5	7	8	6	4

(A) 24 (B) 23.83 (C) 27.16 (D) None of these

MSI080

17. For a normal distribution:

(A) mean = median (B) median = mode

(C) mean = mode (D) mean = median = mode

MSI081

For More Join: @IITJEE_Advanced

18. The range of observations 2, 3, 5, 9, 8, 7, 6, 5, 7, 4, 3 is :

(A) 6 (B) 7 (C) 5.5 (D) 11

MSI082

19. Mean deviation of the observations 70, 42, 63, 34, 44, 54, 55, 46, 38, 48 from median is :

(A) 7.8 (B) 8.6 (C) 7.6 (D) 8.8

MSI083

20. Mean deviation of 5 observations from their mean 3 is 1.2, then the coefficient of mean deviation is:

(A) 0.24 (B) 0.4 (C) 2.5 (D) None of these

MSI084

21. The mean deviation from median is :

(A) greater than the mean deviation from any other central value

(B) less than or equal to the mean deviation from any other central value

(C) equal to the mean deviation from any other central value

(D) maximum if all values are positive

MSI085

22. The S.D. of the first n natural numbers is:

(A) $\sqrt{\frac{n^2-1}{2}}$ (B) $\sqrt{\frac{n^2-1}{3}}$ (C) $\sqrt{\frac{n^2-1}{4}}$ (D) $\sqrt{\frac{n^2-1}{12}}$

MSI086

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23. The variance of observations 112, 116, 120, 125, 132 is :

(A) 58.8 (B) 48.8 (C) 61.8 (D) None of these

MSI087

24. If $\sum_{i=1}^{10}(x_i-15)=12$ and $\sum_{i=1}^{10}(x_i-15)^2=18$, then the S.D. of observations x_1, x_2 ..... x_{10} is :

(A) $\frac{2}{5}$ (B) $\frac{3}{5}$ (C) $\frac{4}{5}$ (D) None of these

MSI088

25. The variance of the series a, $a+d$, $a+2d$, ..., $a+2$nd is :

(A) $\frac{n(n+1)}{2}d^2$ (B) $\frac{n(n+1)}{3}d^2$ (C) $\frac{n(n+1)}{6}d^2$ (D) $\frac{n(n+1)}{12}d^2$

MSI089

26. Variance is independent of change of :

(A) only origin (B) only scale

(C) origin and scale both (D) none of these

MSI090

27. If each observation of a distribution whose variance is σ^2 is multiplied by λ, then the S.D. of the new new observations is :

(A) σ (B) $\lambda\sigma$ (C) $|\lambda|\sigma$ (D) $\lambda^2\sigma$

MSI091

For More Join: @IITJEE_Advanced

28. If $\sum_{i=1}^{11}(x_i-4)=11$ and $\sum_{i=1}^{11}(x_i-4)^2=44$, then find variance of x_1, x_2, x_3 ..... x_{11}.

(A) 4 (B) 3 (C) 7 (D) 11

MSI002

29. If variance of x_1, x_2, x_3, x_4, x_5 is σ^2, find variance of $3x_1+4, 3x_2+4, 3x_3+4, 3x_4+4, 3x_5+4$

(A) $4\sigma^2+3$ (B) $4\sigma^2+9$ (C) $9\sigma^2$ (D) $4\sigma^2-3$

MSI017

30. The mean and variance of 7 observations are 7 and $\frac{100}{7}$. If 5 of the observations are 2, 4, 7, 11, 10, then find the remaining 2 observations.

(A) 3, 6 (B) 3, 12 (C) 4, 11 (D) 5, 10

MSI020

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For More Join: @IITJEE_Advanced

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

1. The average age of a group of men and women is 30 years. If average age of men is 32 and that of women is 27, then the percentage of women in the group is greater than or equal to :

(A) 60 (B) 50 (C) 40 (D) 30

MSI092

2. Mean and median of four numbers a, b, c and d $(b < a < d < c)$ is 35 and 25 respectively, then the value of $b + c - a - d$ will be divisible by :

(A) 8 (B) 5 (C) 40 (D) 20

MSI093

3. The A.M. of first n terms of the series 1.3.5, 3.5.7, 5.7.9,....., is $an^3 + bn^2 + cn + d$, then :

(A) $a + b = 10$ (B) $a - b = 6$ (C) $a + b + c + d = 15$ (D) $a - d = 4$

MSI094

4. The observations 29, 32, 48, 50, x, $x + 2$, 72, 78, 84, 95 are arranged in ascending order and their median is 63, then the value of x is less than :

(A) 61 (B) 62 (C) 62.5 (D) 63

MSI095

5. If the mode of the distribution is 18 and the mean is 24, then median is greater than :

(A) 18 (B) 24 (C) 22 (D) 21

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MSI096

6. For 15 observations of x, mean and median were found to be 12 and 20 respectively. Later an observation which was 25 found to be wrong then replaced by its correct value 55, if new mean and median are a and b respectively, then :

(A) $a + b = 65$ (B) $ab = 280$ (C) $a - b = -6$ (D) $b = a + 6$

MSI097

7. The mean square deviation about –1 and +1 of a set of observations are 7 and 3 respectively, then standard deviation of the set is greater than or equal to :

(A) $\sqrt{2}$ (B) $\sqrt{3}$ (C) 2 (D) 1

MSI098

8. The S.D. of the following frequency distribution is greater than or equal to :

Class	0-10	10-20	20-30	30-40
f_i	1	3	4	2

(A) 7.8 (B) 9 (C) 8.1 (D) 0.9

MSI099

9. The S.D. of the numbers 31, 32, 33,.... 47 is less than or equal to :

(A) $2\sqrt{6}$ (B) $4\sqrt{3}$ (C) $\sqrt{\frac{47^2-1}{12}}$ (D) $2\sqrt{5}$

MSI100

For More Join: @IITJEE_Advanced

10. The median and standard deviation (S.D.) of a distribution will be, if each term is increased by 2 :
(A) S.D. will increased by 2 (B) median will increased by 2
(C) median will remain same (D) S.D. will remain same

MSI101

11. The coefficient of mean deviation from median of observations 40, 62, 54, 90, 68, 76 is less than or equal to :
(A) 2.16 (B) 0.2 (C) 5 (D) 0.1

MSI102

12. A group of 10 observations has mean 5 and S.D. $2\sqrt{6}$. Another group of 20 observations has mean 5 and S.D. $3\sqrt{2}$, then the S.D. of combined group of 30 observations is less than or equal to :
(A) $\sqrt{5}$ (B) $2\sqrt{5}$ (C) $3\sqrt{5}$ (D) $4\sqrt{5}$

MSI103

13. x_i 's are observations such that $(x_i^2 - 7x_i + 12)(x_i^3 - x_i^2 - 4x_i + 4) = 0$, then :
(A) Mean = $\frac{8}{5}$ (B) Median = 2 (C) Variance = $\frac{106}{25}$ (D) Mean = 2

MSI104

14. Which of the following is/are correct ?
(A) $\sigma^2 = \frac{\sum_{i=1}^{n}(x_i-1)^2}{n} - \left(\frac{\sum_{i=1}^{n}(x_i-1)}{n}\right)^2$ (B) $\sigma = \frac{\sum_{i=1}^{n}x_i^2}{n} - \left(\frac{\sum_{i=1}^{n}x_i}{n}\right)^2$
(C) $\sigma^2 = \frac{\sum_{i=1}^{n} f_i(x_i-\bar{x})^2}{\sum_{i=1}^{n} f_i}$ (D) $\sigma^2 = \frac{\sum_{i=1}^{n} f_i(x_i-k)^2}{\sum_{i=1}^{n} f_i} - (\bar{x})^2$

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MSI105

15. Which of the following can not be obtained from graph ?
(A) Mean (B) Median (C) Mode (D) Variance

MSI106

COMPREHENSION TYPE QUESTIONS

Paragraph for Question 16 and 17

If the variance of $x_1, x_2, ..., x_n$ is σ^2 then
var. of $x_1 \pm \lambda, x_2 \pm \lambda, ..., x_n \pm \lambda$ is σ^2 and var. of $\lambda x_1, \lambda x_2, ..., \lambda x_n$ is $\lambda^2\sigma^2$

16. If each observation of a distribution whose S.D. is σ, is increased by λ, then the variance of the new observations is :
(A) σ (B) $\sigma + \lambda$ (C) σ^2 (D) $\sigma^2 + \lambda$

MSI107

17. The standard deviation of variate x_i is σ. Then standard deviation of the variate $\frac{ax_i + b}{c}$, where a, b, c are constants is :
(A) $\left(\frac{a}{c}\right)\sigma$ (B) $\left|\frac{a}{c}\right|\sigma$ (C) $\left(\frac{a^2}{c^2}\right)\sigma$ (D) None of these

MSI108

For More Join: @IITJEE_Advanced

Paragraph for Question 18 to 20

If the mean and variance of the frequency distribution

x_i	2	4	6	8	10	12	14	16
f_i	4	4	α	15	8	β	4	5

are 9 and 15.08 respectively, then

18. Then α + β is divisible by :

(A) 10 (B) 2 (C) 5 (D) 15

MSI109

19. Then equation $x^2 - (\alpha + \beta + 1)x + \alpha\beta + \alpha = 0$ has roots :

(A) 5 (B) –4 (C) 6 (D) –5

MSI110

20. Mean of αβ, 8, 10, $\alpha^2 + \beta$, 2, 6 is less than :

(A) 14 (B) 12 (C) 15 (D) 18

MSI111

Paragraph for Question 21 and 22

A student noted the number of cars passing through a spot on a road for 100 periods each of 3 minutes and summarised it in the table given below

Number of cars	Frequence
0–10	7
10–20	14
20–30	13
30–40	12
40–50	20
50–60	11
60–70	15
70–80	8

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21. Find mean of the data.

MSI112

22. Find mode of the data.

MSI113

For More Join: @IITJEE_Advanced

MATCHING LIST TYPE QUESTION

Answer Q.23 and Q.24 by appropriately matching the information given in the two List of the following table

	List-I		**List-II**
(I)	The weighted mean of first n natural numbers whose weights are equal is :	(P)	$\frac{n^2-1}{3}$
(II)	Variance of the group $\alpha, \alpha+2, \alpha+4, \alpha+6, \ldots\ldots$ upto n terms ($\alpha \neq 0$) is :	(Q)	$\frac{n}{2}$
(III)	The mean of values $0, 1, 2, \ldots\ldots, n$ when their weights are $1, {}^nC_1, {}^nC_2, \ldots., {}^nC_n$, respectively is :	(R)	$\frac{n-1}{2}$
(IV)	The mean of n values of a distribution is $\bar{x}$. If its first value is increased by 1, second by 2, ... then the mean of new values will increase by	(S)	$\frac{n+1}{3}$
		(T)	$\frac{n+1}{2}$

23. Which of the following is the only CORRECT combination ?

(A) (I), (S) (B) (II), (Q) (C) (I), (T) (D) (IV), (Q)

MSI114

24. Which of the following is the only CORRECT combination ?

(A) (I), (P) (B) (II), (T) (C) (III), (S) (D) (IV), (T)

MSI115

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MATRIX MATCH TYPE QUESTION

25. Consider the following distribution of profits earned by 30 shops.

Profit (Rs. In Lakhs)	Number of Shope
1–10	2
10–20	10
20–30	7
30–40	7
40–50	4

	Column-I		**Column-II**
(A)	Mean	(P)	17.27
(B)	Mode	(Q)	136.73
(C)	Median	(R)	25.33
(D)	Variance	(S)	24.28

MSI116

For More Join: @IITJEE_Advanced

EXERCISE (S)

1. If the mean of following frequency distribution is 2.6, then the value of f is :-

x_i	1	2	3	4	5
f_i	5	4	f	2	3

MSI117

2. If the coefficient of variation and standard deviation of a distribution are 50% and 20 respectively, then its mean is-

MSI118

3. The mean deviation of the numbers 1, 2, 3, 4, 5 is-

MSI119

4. The median of 21 observations is 40. If each observation greater than the median are increased by 6, then the median of the observations will be-

MSI120

5. The mean of a distribution is 4. If its coefficient of variation is 58%, then the S.D. of the distribution is:-

MSI121

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6. The average age of a teacher and three students is 20 years. If all students are of equal age and the difference between the age of the teacher and that of a student is 20 years, then the age (in years) of the teacher is-

MSI122

7. The mean deviation from mean of observations 5, 10, 15, 20, ...85 is :-

MSI123

8. If standard deviation of variate x_i is 10, then variance of the variate $(50 + 5x_i)$ will be-

MSI124

9. The median of 19 observations of a group is 30. If two observations with values 8 and 32 are further included, then the median of the new group of 21 observation will be

MSI125

10. Find the median of 5 observations i.e. $3^{\log_9 4}, 5^{\log_{1/2} 8}, e^{2\ell n 3}, \ell n\left(\frac{1}{e^2}\right)+3, e^{2\ell n 3+\frac{1}{\log_4 e}}$.

MSI126

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

1. If the standard deviation of the numbers 2, 3, a and 11 is 3.5, then which of the following is true ? **[JEE (Main) 2016]**

(1) $3a^2 - 23a + 44 = 0$ (2) $3a^2 - 26a + 55 = 0$

(3) $3a^2 - 32a + 84 = 0$ (4) $3a^2 - 34a + 91 = 0$

MSI127

2. If $\sum_{i=1}^{9}(x_i - 5) = 9$ and $\sum_{i=1}^{9}(x_i - 5)^2 = 45$, then the standard deviation of the 9 items $x_1, x_2, ..., x_9$ is - **[JEE (Main) 2018]**

(1) 4 (2) 2 (3) 3 (4) 9

MSI128

3. 5 students of a class have an average height of 150 cm and variance 18 cm^2. A new student, whose height is 156 cm joined them. The variance (in cm^2) of the height of these six students is: **[JEE (Main) 2019]**

(1) 22 (2) 20 (3) 16 (4) 18

MSI129

4. The mean of five observations is 5 and their variance is 9.20. If three of the given five observations are 1, 3 and 8, then the ratio of other two observations is : **[JEE (Main) 2019]**

(1) 4 : 9 (2) 6 : 7 (3) 5 : 8 (4) 10 : 3

MSI130

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5. The mean and the variance of five observation are 4 and 5.20 respectively. If three of the observations are 3, 4 and 4, then absolute value of the difference of the other two observations is : **[JEE (Main) 2019]**

(1) 1 (2) 3 (3) 7 (4) 5

MSI131

6. The mean and variance of seven observations are 8 and 16 respectively. If 5 of the observations are 2, 4, 10, 12, 14, then the product of the remaining two observations is : **[JEE (Main) 2019]**

(1) 40 (2) 49 (3) 48 (4) 45

MSI132

7. If for some $x \in R$, the frequency distribution of the marks obtained by 20 students in a test is :

Marks	2	3	5	7
Frequency	$(x+1)^2$	$2x+5$	x^2-3x	x

then the mean of the marks is : **[JEE (Main) 2019]**

(1) 2.8 (2) 3.2 (3) 3.0 (4) 2.5

MSI133

8. If both the mean and the standard deviation of 50 observations $x_1, x_2, ..., x_{50}$ are equal to 16, then the mean of $(x_1 - 4)^2, (x_2 - 4)^2, ...(x_{50} - 4)^2$ is : **[JEE (Main) 2019]**

(1) 525 (2) 380 (3) 480 (4) 400

MSI134

For More Join: @IITJEE_Advanced

9. The mean and the standard deviation (s.d.) of 10 observations are 20 and 2 respectively. Each of these 10 observations are multiplied by p and then reduced by q, where $p \neq 0$ and $q \neq 0$. If the new mean and new s.d. become half of their original values, then q is equal to **[JEE (Main) 2020]**

(1) –20 (2) 10 (3) –10 (4) –5

MSI135

10. Let the observations $x_i (1 \le i \le 10)$ satisfy the equations, $\sum_{i=1}^{10}(x_i - 5) = 10$ and $\sum_{i=1}^{10}(x_i - 5)^2 = 40$.

If μ and λ are the mean and the variance of the observations, $x_1 - 3$, $x_2 - 3$, ..., $x_{10} - 3$, then the ordered pair (μ, λ) is equal to: **[JEE (Main) 2020]**

(1) (6, 6) (2) (3, 6) (3) (6, 3) (4) (3, 3)

MSI136

11. Consider three observations a, b and c such that $b = a + c$. If the standard deviation of $a + 2$, $b + 2$, $c + 2$ is d, then which of the following is true ? **[JEE (Main) 2021]**

(1) $b^2 = 3(a^2 + c^2) + 9d^2$ (2) $b^2 = a^2 + c^2 + 3d^2$

(3) $b^2 = 3(a^2 + c^2 + d^2)$ (4) $b^2 = 3(a^2 + c^2) - 9d^2$

MSI137

12. If the variance of 10 natural numbers 1, 1, 1,...., 1, k is less than 10, then the maximum possible value of k is _____. **[JEE (Main) 2021]**

MSI138

13. Let $X_1, X_2, ..., X_{18}$ be eighteen observations such that $\sum_{i=1}^{18}(X_i - \alpha) = 36$ and $\sum_{i=1}^{18}(X_i - \beta)^2 = 90$, where α and β are distinct real numbers. If the standard deviation of these observations is 1, then the value of $|\alpha - \beta|$ is _____. **[JEE (Main) 2021]**

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MSI139

14. Consider the statistics of two sets of observations as follows :

	Size	Mean	Variance
Observation I	10	2	2
Observation II	n	3	1

If the variance of the combined set of these two observations is $\frac{17}{9}$, then the value of n is equal to _____. **[JEE (Main) 2021]**

MSI140

15. Consider a set of $3n$ numbers having variance 4. In this set, the mean of first $2n$ numbers is 6 and the mean of the remaining n numbers is 3. A new set is constructed by adding 1 into each of first $2n$ numbers and subtracting 1 from each of the remaining n numbers. If the variance of the new set is k, then $9k$ is equal to _____. **[JEE (Main) 2021]**

MSI141

16. The mean and standard deviation of 50 observations are 15 and 2 respectively. It was found that one incorrect observation was taken such that the sum of correct and incorrect observations is 70. If the correct mean is 16, then the correct variance is equal to : **[JEE (Main) 2022]**

(1) 10 (2) 36 (3) 43 (4) 60

MSI142

For More Join: @IITJEE_Advanced

17. The mean of the numbers a, b, 8, 5, 10 is 6 and their variance is 6.8. If M is the mean deviation of the numbers about the mean, then 25 M is equal to : **[JEE (Main) 2022]**

(1) 60 (2) 55 (3) 50 (4) 45

MSI143

18. The number of values of $a \in \mathbb{N}$ such that the variance of 3, 7, 12 a, 43 – a is a natural number is:

[JEE (Main) 2022]

(1) 0 (2) 2 (3) 5 (4) infinite

MSI144

19. Let the mean and the variance of 5 observations x_1, x_2, x_3, x_4, x_5 be $\frac{24}{5}$ and $\frac{194}{25}$ respectively. If the mean and variance of the first 4 observations are $\frac{7}{2}$ and a respectively, then $(4a + x_5)$ is equal to: **[JEE (Main) 2022]**

(1) 13 (2) 15 (3) 17 (4) 18

MSI145

20. Let $9 = x_1 < x_2 < ... < x_7$ be in an A.P. with common difference d. If the standard deviation of $x_1, x_2 ..., x_7$ is 4 and the mean is $\bar{x}$, then $\bar{x} + x_6$ is equal to : **[JEE (Main) 2023]**

(1) $18\left(1+\frac{1}{\sqrt{3}}\right)$ (2) 34 (3) $2\left(9+\frac{8}{\sqrt{7}}\right)$ (4) 25

MSI057

21. Let the mean and variance of 12 observations be $\frac{9}{2}$ and 4 respectively. Later on, it was observed that two observations were considered as 9 and 10 instead of 7 and 14 respectively. If the correct variance is $\frac{m}{n}$, where m and n are co-prime, then $m + n$ is equal to **[JEE (Main) 2023]**

(1) 316 (2) 314 (3) 317 (4) 315

MSI060

22. If the mean and variance of the frequency distribution

x_i	2	4	6	8	10	12	14	16
f_i	4	4	α	15	8	β	4	5

are 9 and 15.08 respectively, then the value of $\alpha^2 + \beta^2 - \alpha\beta$ is. **[JEE (Main) 2023]**

MSI062

23. Let μ be the mean and σ be the standard deviation of the distribution

x_i	0	1	2	3	4	5
f_i	$k+2$	$2k$	k^2-1	k^2-1	k^2+1	$k-3$

where $\sum f_i = 62$. If $[x]$ denotes the greatest integer $\le x$, then $[\mu^2 + \sigma^2]$ is equal **[JEE (Main) 2023]**

(1) 8 (2) 7 (3) 6 (4) 9

MSI063

24. Let $X = \{11, 12, 13, ..., 40, 41\}$ and $Y = \{61, 62, 63, ..., 90, 91\}$ be the two sets of observations. If $\bar{x}$ and $\bar{y}$ are their respective means and σ^2 is the variance of all the observations in $X \cup Y$, then $|\bar{x} + \bar{y} - \sigma^2|$ is equal to. **[JEE (Main) 2023]**

MSI064

For More Join: @IITJEE_Advanced

25. Let the mean and standard deviation of marks of class A of 100 students be respectively 40 and $\alpha(>0)$, and the mean and standard deviation of marks of class B of n students be respectively 55 and $30-\alpha$. If the mean and variance of the marks of the combined class of $100+n$ students are respectively 50 and 350, then the sum of variances of classes A and B is: **[JEE (Main) 2023]**

(1) 500 (2) 650 (3) 450 (4) 900

MSI065

26. Consider 10 observations $x_1, x_2, \ldots, x_{10}$. such that $\sum_{i=1}^{10}(x_i-a) = 2$ and $\sum_{i=1}^{10}(x_i-\beta)^2 = 40$, where α, β are positive integers. Let the mean and the variance of the observations be $\frac{6}{5}$ and $\frac{84}{25}$ respectively. Then $\frac{\beta}{\alpha}$ is equal to : **[JEE (Main) 2024]**

(1) 2 (2) $\frac{3}{2}$ (3) $\frac{5}{2}$ (4) 1

MSI146

27. Let $a, b, c \in N$ and $a < b < c$. Let the mean, the mean deviation about the mean and the variance of the 5 observations 9, 25, a, b, c be 18, 4 and $\frac{136}{5}$, respectively. Then $2a + b - c$ is equal to _____ .

[JEE (Main) 2024]

MSI147

28. Let the median and the mean deviation about the median of 7 observations 170, 125, 230, 190, 210, a, b be 170 and $\frac{205}{7}$ respectively. Then the mean deviation about the mean of these 7 observations is : **[JEE (Main) 2024]**

(1) 31 (2) 28 (3) 30 (4) 32

MSI148

29. The mean and standard deviation of 15 observations were found to be 12 and 3 respectively. On rechecking it was found that an observation was read as 10 in place of 12. If μ and σ^2 denotes the mean and variance of the correct observations respectively, then $15(\mu+\mu^2+\sigma^2)$ is equal to : **[JEE (Main) 2024]**

MSI149

30. If the variance of the frequency distribution is 160, then the value of $c \in N$ is

x	c	$2c$	$3c$	$4c$	$5c$	$6c$
f	2	1	1	1	1	1

[JEE (Main) 2024]

(1) 5 (2) 8 (3) 7 (4) 6

MSI150

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EXERCISE - JEE (Advanced) PYQ

1. Consider the given data with frequency distribution

x_i	3	8	11	10	5	4
f_i	5	2	3	2	4	4

Match each entry in **List-I** to the correct entries in **List-II**.

List-I		**List-II**	
(P)	The mean of the above data is	(1)	2.5
(Q)	The median of the above data is	(2)	5
(R)	The mean deviation about the mean of the above data is	(3)	6
(S)	The mean deviation about the median of the above data is	(4)	2.7
		(5)	2.4

The correct option is: **[JEE (Advanced) 2023]**

(A) (P) → (3) (Q) → (2) (R) → (4) (S) → (5)

(B) (P) → (3) (Q) → (2) (R) → (1) (S) → (5)

(C) (P) → (2) (Q) → (3) (R) → (4) (S) → (1)

(D) (P) → (3) (Q) → (3) (R) → (5) (S) → (5)

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MSI151

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ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	B	B	D	D	B	C	D	B	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B	C	D	C	B	B	D	B	B	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B	D	B	B	B	A	C	B	C	B

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C,D	A,B,C,D	A,C,D	C,D	A,D	B,C,D	A,B,D	A,B,C,D	A,B,C	B,D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A,B,C	B,C,D	B,C,D	A,C	A,D	C	B	A,B,C	A,C	A,C,D
Que.	21	22	23	24	25					
Ans.	40.70	44.70	C	D	(A → R; B → P; C → S; D → Q)					

EXERCISE (S)

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1. 1 **2.** 40 **3.** 1.20 **4.** 40 **5.** 2.32
6. 35 **7.** 21.17 **8.** 2500 **9.** 30.00 **10.** 9.00

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	2	2	1	3	3	1	4	1	4
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	4	11	4	5	68	3	1	1	2	2
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	3	25	1	603	1	1	33	3	2521	3

EXERCISE - JEE (Advanced) PYQ

Que.	1
Ans.	A

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Important Notes

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